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Technical Change and the Rate of Profit in Classical-Marxian Models of Economic Growth

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Abstract

I study the effect of viable technical change on the equilibrium profit rate in Classical-Marxian models of economic growth with alternative labor market closures. Capitalists adopt a new technique of production only if it is expected to increase the profit rate at the existing real wage rate. I consider three alternative closures: (a) constant real wage rate (labor surplus economy), (b) constant wage share (advanced capitalist economy with strong labor), and (c) constant unemployment rate (advanced capitalist economy with weak labor). I show that the equilibrium profit rate can unambiguously fall after viable capital-using, labor-saving (CU-LS) technical for an advanced economy with strong labor.

JEL Classification: B51, C02

Keywords

economic growth, technical change, falling rate of profit

1. Introduction

The impact of technical change on the profit rate has been an important question in classical political economy. For David Ricardo, resource constraints would push a capitalist economy toward a stationary state of zero growth in the long run, unless technical progress, which increased the productivity of labor and capital, came to the rescue.

Karl Marx, a dialectical thinker par excellence, presented a more contradictory picture of technical progress in capitalism (Marx 1993). Technical change, Marx argued, is at one and the same time capitalism's greatest strength and weakness. While technical change makes the capitalist system progressive and opens up possibilities of material progress, it can also push the system toward stagnation by driving the profit rate down. If the profit rate falls below the minimum threshold level necessary to induce capitalists to undertake investment, capital accumulation, the very engine of growth in capitalism, can fall to zero.

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Marx's argument about the law of tendential fall in the rate of profit (LTFRP) was simple, straightforward, and powerful. It can be presented in the language of contemporary macroeconomics as follows: Technical change in capitalism is characterized by the unrelenting march of mechanization, of labor being replaced by capital; that is, technical change is labor saving. While this will increase labor productivity (the ratio of output and labor), it can also reduce capital productivity (the ratio of output and the capital stock); that is, technical change can, in addition, be capital using. Such a pattern of capital-using labor-saving (CU-LS) technical change, which seems to be the norm in capitalist economies for extended periods of time and which has been termed Marx-biased technical change (MBTC) by Foley et al. (2019), will impart a tendency to the equilibrium profit rate to fall *if the rate of exploitation, or equivalently the profit share in national income, remains relatively stable*. This is because the rate of profit is the product of the profit share and capital productivity.

While many generations of scholars and activists debated Marx's LTFRP since the publication of volume 3 of *Capital*, it was Nobuo Okishio's contribution that propelled the literature in a whole new direction. Okishio (1961) argued that technical change occurs in capitalist economies when capitalist firms adopt new techniques of production. But capitalist firms will adopt a new technique of production only when it reduces the cost of production at the prevailing prices and wages. We can call such a new technique of production as (capitalistically) *viable*. Okishio (1961) demonstrated, in a multisector model with linear technologies of production, that the equilibrium profit rate will rise, and not fall, with viable technical change if the real wage rate remains constant.

Following Okishio's insightful and provocative analysis, a large body of literature has studied different theoretical aspects of the question of the evolution of the rate of profit after viable technical change (Laibman 1982; Foley 1986; Skott 1992; Skillman 1997; Negishi 1998; Tavani and Zamparelli 2017; Blecker and Setterfield 2019; Basu 2021; Basu and Orellana 2023). What is relevant to this study is a specific issue that has often not been adequately highlighted in the extant literature: The *trajectory of the real wage rate* after technical change is key in determining the trajectory of the equilibrium profit rate.

Since the movement of the real wage rate will depend on the specific structure of the labor market or the broader institutional setup for capital-labor bargaining, the LTFRP literature can be enriched further by paying closer attention to the context provided by the specific structure of the economy. After all a labor surplus developing economy undergoing capitalist development might work with a different logic, compared to a developed capitalist economy, so far as the impact of technical change on profitability is concerned precisely because the structure of the labor markets are so different in the two. Such a perspective, pre-dating the Okishio-inspired debates by several decades, seems to have informed Maurice Dobb's comments on Marx's LTFRP (Dobb 1945: chapter iv).

In this article, I revisit Marx's LTFRP by studying the impact of viable technical change on the equilibrium rate of profit in classical-Marxian models of economic growth presented in Blecker and Setterfield (2019: section 2.4.3). The basic model has the following features: (a) Leontief technology of production; (b) standard choice of technique by capitalist firms based on profit rate comparisons before and after technical change; (c) an accumulation (saving/investment) function with a minimum threshold for positive investment; and (d) a fixed, normal (or desired) capacity utilization in the long run.

I connect with Maurice Dobb's useful analysis by working with alternative labor market closures of the basic classical-Marxian growth model.[1] My interpretation of these alternative closures maps them into different structural and institutional settings of capitalist economies thereby allowing for *alternative trajectories of the real wage rate*: a labor surplus developing economy,

[1] This strategy of using alternative closures of classical models of economic growth can also be seen in Kaldor (1989).

an advanced capitalist economy with strong labor, and an advanced capitalist economy with weak labor.

Through my theoretical analysis, I find the following: (a) the equilibrium profit rate will rise after viable, CU-LS technical change in a labor surplus economy; (b) the equilibrium profit rate will fall after viable, CU-LS technical change in an advanced capitalist economy with strong labor; (c) the equilibrium profit rate will rise after viable, CU-LS technical change in an advanced capitalist economy with weak labor if the labor-saving aspect of the technical change is weak; it *can* fall if the labor-saving aspect of the technical change is strong. Thus, within the context of one-sector classical-Marxian growth models, the prospect of Marx's LTFRP holding is limited—it holds unambiguously only in one specific type of labor market arrangement, where labor is powerful enough to maintain a constant wage share in national income.

The rest of this article is organized as follows: in section 2, I present a classical-Marxian model of economic growth; in section 3, I discuss the issue of choice of technique by capitalists firms; in section 4, I discuss the main question of this study about the movement of the equilibrium profit rate after viable technical change; I conclude the article in section 5. Some proofs are collected in the appendix.

2. Classical-Marxian Growth Models With Alternative Closures

2.1. The basic model

2.1.1. Production and technology. I consider a one-sector closed capitalist economy without government. Let Y, N, L, and K denote real output, the labor force, level of employment, and real capital stock, respectively. Production is characterized by a constant coefficient Leontief technology, defined as follows:

$$Y = \min\left(\frac{L}{a_0}, \frac{K}{a_1}\right) \tag{1}$$

where a_0 denotes the reciprocal of labor productivity (Y/L) and a_1 represents the reciprocal of capital productivity (Y/K). For a given configuration of technology, a_0 and a_1 are strictly positive and constant. Since I am interested in studying long-run dynamics, I assume a constant capacity utilization rate and normalize it to unity.

There are two important trade-offs in this economy, the first between wages and profits, and the second between growth and consumption. The first trade-off is represented by a wage-profit frontier, and the second is represented by a consumption-growth frontier.

2.1.2. Wage-profit frontier. Looked at from the income side, the total value of output is equal to the sum of wage income and profit income:

$$PY = WL + rPK$$

where P, W, and r denote the price level, the nominal wage rate, and the profit rate, respectively. Dividing through by PY and denoting the real wage rate by $w = W/P$, we get the wage-profit frontier:

$$w = \frac{1}{a_0} - \frac{a_1}{a_0} r \tag{2}$$

Since $a_1/(a_0) > 0$, this gives us a negative relationship between the real wage rate and the profit rate. The wage-profit frontier captures the conflict of interest between the workers (who earn

wage income) and the capitalists (who earn profit income) conditional on technology (captured by a_0 and a_1).

2.1.3. Consumption-growth frontier. Looked at from the expenditure side, the total value of output is equal to the sum of consumption and investment expenditure:

$$PY = PC + PI$$

where C, I denote total real consumption and investment expenditure, respectively. Dividing through by PY and denoting the growth rate of capital stock by $g = \Delta K / K = I / K$, we get the consumption-growth frontier:

$$c = \frac{1}{a_0} - \frac{a_1}{a_0} g \tag{3}$$

where c represents real consumption per employed worker. Since $a_1 / (a_0) > 0$, this gives us a negative relationship between consumption and growth. For a given configuration of technology given by a_0 and a_1, the growth rate of capital can be increased only by reducing consumption and increasing accumulation.

2.1.4. Accumulation function. In the classical-Marxian vision, capital accumulation is driven by capitalists, who accumulate (i.e., save and invest) a constant fraction of their profit income in augmenting the capital stock. This is captured by an accumulation function:

$$g = \begin{cases} s_r (r - r_m), & \text{if } r > r_m \geq 0 \\ 0, & \text{if } r \leq r_m \end{cases} \tag{4}$$

where g is the rate of growth of the capital stock and r_m denotes the minimum profit rate that is necessary to induce capitalist firms to undertake positive investment.[2] If the actual profit rate falls to or below r_m, capitalists consume all of their profit income and do not invest. When the profit rate rises above r_m, capitalists invest a fraction, s_r, of their profit income, where $0 < s_r < 1$.[3] The fraction s_r can be called the *accumulation propensity* of capitalists.

What I have outlined above is the basic classical-Marxian growth model. It has four endogenous variables: real wage rate, w; consumption per worker, c; growth rate of capital stock, g; and the profit rate, r. The following exogenous variables are used in the model: a_0, a_1, s_r, r_m. The endogenous variables are related through the following three equations explained above: (2), (3) and (4). Thus, we need at least one more equation to close and solve the model. This allows us to use alternative closures of the model.

2.2. Alternative labor market closures

2.2.1. Labor surplus developing economies. The first closure I consider is where the real wage rate remains unchanged:

$$w = \bar{w} \tag{5}$$

[2] I ignore depreciation of the capital stock. Hence, $g = \Delta K/K = I/K = S/K$, where I is investment and S is savings.

[3] When $r > r_m \geq 0$, total profit income is $rPK = (r - r_m)PK + r_m PK$. Capitalists invest $s_r (r - r_m) PK$ and consume the rest, $(1 - s_r)(r - r_m) PK + r_m PK$.

and where $\bar{w}$ is a constant. This closure is relevant for a labor surplus economy, like one studied extensively in classical development economics (Lewis 1954); it can also be understood as a developing economy undergoing capitalist development through what Marx called primitive accumulation (Dobb 1945). This assumption was used in the "first" classical model of economic growth with unlimited supply of labor in Kaldor (1989).

Drawing on insights from classical development economics, we can characterize a labor surplus economy as being composed of two sectors—a modern, capitalist sector and a traditional, non-capitalist sector. The latter is populated by a very large number of small-scale peasant producers working on their farms with family labor and generating a low level of income. As long as $\bar{w}$ is above the average peasant income, the capitalist sector can draw out the surplus labor from the traditional sector with very little pressure on wages and without reducing agricultural output. This feature is captured by the constancy of the real wage rate in (5).

To summarize, the classical-Marxian growth model for a labor surplus developing economy has four endogenous variables, w, c, g, and r, related to each other by four equations: (2), (3), (4), and (5). To ensure meaningful equilibrium solutions of this model, we need the following restrictions.

Assumption 1. The *parameters* $a_0, a_1, r_m, \bar{w}$ *are strictly positive,* $0 < s_r < 1$, *and the feasible region of technological possibilities is given by:*

$$a_0 < \frac{1}{\bar{w}},\ a_1 < \frac{1}{r_m} - \left(\frac{\bar{w}}{r_m}\right) a_0 \tag{6}$$

The first part of the assumption states that labor productivity, $1/a_0$, must be larger than the fixed real wage rate, $\bar{w}$; if labor productivity fell below this level, then capitalist production would not be feasible because output would be lower than the wage rate (so that profit income would be negative). The second part gives a lower bound for the capital productivity, $1/a_1$, that keeps the profit rate above the minimum threshold, r_m. If capital productivity fell below this level, given a_0, then the profit rate would fall below the threshold and the economy would grind to a halt.

2.2.2. Advanced capitalist countries with strong labor. The second closure I consider is where the wage share (ratio of wage income to total income) or equivalently the profit share (ratio of profit income to total income) remains constant:

$$\pi = \bar{\pi} \tag{7}$$

and where $0 < \bar{\pi} < 1$ is a constant. This model has been called the *conventional wage share model* by Foley et al. (2019). It is applicable to an advanced capitalist economy that is no longer characterized by surplus labor and, in addition, where labor is powerful enough to force the real wage rate to grow in tandem with labor productivity in the long run.

What dynamic process of adjustment ensures a constant profit (or wage) share? We can think of the underlying process as involving collective action by workers to enforce a certain (functional) distribution of national income. To be concrete, we can think of workers as having a target wage share, η, that they want to, and have the power to, attain. When the actual wage share, $a_0 w$, falls below the target wage share, they push the real wage rate up through collective action; when the actual wage share moves above the target wage share, workers let up on collective action and allow the real wage to fall. When the actual and target wage shares coincide, workers do not take any action and the real wage rate stays put.

This adjustment process can be captured by the function $\psi(.)$ in (8), which specifies the rate of change of the real wage rate:

$$\dot{w} = \psi\left(a_0 w - \eta\right), \psi\left(0\right) = 0, \psi'(.) < 0 \tag{8}$$

When the actual wage share is below the target wage share, $a_0 w - \eta < 0$, the real wage rate rises, $\dot{w} > 0$; when the actual wage share is above the target wage share, $a_0 w - \eta > 0$, the real wage rate falls, $\dot{w} > 0$; when the two are equal, there is no change in the real wage rate, $\dot{w} > 0$. This last condition defines a position of equilibrium, and it is easy to see that the unique equilibrium for the one-dimensional dynamical system in (8) is stable because:

$$\frac{d\dot{w}}{dw} = \psi'(.) \times a_0 < 0$$

Moreover, we can see that the adjustment process will ensure a stable profit share given by:

$$\bar{\pi} = 1 - \eta$$

which is what figures in (7).

In advanced capitalist countries, the wage share has remained constant for long periods of time (running right up to the end of the 1970s). In the early 1960s, this led Nicholas Kaldor to present this as one of the "stylized facts" of economic growth in capitalism (Kaldor 1989). The classical conventional wage share model incorporates this stylized fact. Therefore, it remains an important version of the classical-Marxian growth model to study advanced capitalist countries.

To summarize, the classical-Marxian growth model for an advanced capitalist country with strong labor has four endogenous variables, w, c, g, and r, related to each other by four equations, (2), (3), (4), and (7). I need to impose the following restrictions to ensure meaningful equilibrium solutions of this model.

Assumption 2. *The parameters* a_0, a_1, r_m *are strictly positive;* $0 < \bar{\pi} < 1,\ 0 < s_r < 1$, *and the feasible region of technological possibilities is given by:*

$$a_1 < \frac{\bar{\pi}}{r_m} \tag{9}$$

This assumption gives a lower bound for the capital productivity, $1/a_1$, that keeps the profit rate above the minimum threshold, r_m. If capital productivity fell below this level, given a_0, then the profit rate would fall below the threshold and the economy would stop growing. This assumption rules that out.

2.2.3. Advanced capitalist countries with weak labor. There is a growing body of evidence which shows that the pattern of constant wage share, which had been observed for many decades, changed in the early 1980s. Since then the wage share has declined, in both advanced capitalist and developing capitalist economies (IMF 2017). Political and institutional factors that had forced real wages to grow at the same rate as labor productivity seems to have been weakened by technological change, globalization of production, decline in unionization rates and other such factors. Hence, it seems theoretically desirable to allow for closures of the classical-Marxian growth model where the wage share is not held constant *as an assumption*. An alternative stylized fact might recommend itself: Over the long run, the unemployment rate is constant. We can capture this assumption and generate another closure of the model by equating the growth rates of the supply of and demand for labor.

Labor supply: Following Marx, I assume that in capitalist economies, labor supply is socially determined. At the very least, labor supply can be augmented by labor-saving technical change and by drawing latent reserves of labor (within the household or in small-scale agriculture) into the orbit of profit-oriented commodity production. Both processes of labor supply growth are impacted by the real wage rate. On the one hand, real wage pressures induce capitalists to search for and adopt labor-saving technical change; on the other hand, high real wage rates increase the incentive to supply household and petty-commodity production labor to the capitalist labor market. Hence, a non-negative response of labor supply growth to the real wage rate seems a natural way to model these features of capitalist labor markets:

$$n = n_0 + n_1 w \tag{10}$$

where n is the growth rate of labor supply, $n_0 > 0$ represents the base growth rate (driven by population growth rate, immigration, etc.), and $n_1 \geq 0$ captures the response of labor supply growth to real wage increases.[4] Different values of n_1 can be used to capture different subcases, which I comment on below.

Labor demand: Recall that for a given configuration of technology, a_0 (reciprocal of labor productivity) and a_1 (reciprocal of capital productivity) are constant. Hence, the growth rate of capital stock is equal to the growth rate of output, which, in turn, is equal to the growth rate of the demand for labor. These three growth rates are given by g. On the other hand, the growth rate of the supply of labor is given by n in (10), which responds to the level of the real wage rate, w.

Adjustment and equilibrium: The real wage rate is the adjustment variable, which changes to establish equilibrium in the labor market in the long run. The rate of change of the real wage rate responds positively to the difference between the growth rates of the demand for and supply of labor:

$$\dot{w} = \phi(g - n),\ \phi(0) = 0, \phi'(.) > 0 \tag{11}$$

and a steady-state, long-run equilibrium of the model is obtained when $\dot{w} = 0$:

$$g = n \tag{12}$$

Two things about the equilibrium should be noted. First, the equality of the growth rates of labor demand and supply does not imply full employment in equilibrium; it only implies that the unemployment rate will remain constant in equilibrium. For instance, if there was unemployment in the initial period, then the equilibrium unemployment rate will be the same as that rate. Second, the unique equilibrium of the one-dimensional dynamical system in (11) is stable because:

$$\frac{d\dot{w}}{dw} = \phi'(.) \times \left(\frac{dg}{dw} - \frac{dn}{dw} \right) = \phi'(.) \times \left(-\frac{s_r a_0}{a_1} - n_1 \right) < 0$$

where I have used:

$$g = s_r r - s_r r_m = s_r \left(\frac{1 - a_0 w}{a_1} \right) - s_r r_m = \left(\frac{s_r}{a_1} - s_r r_m \right) - \left(\frac{s_r a_0}{a_1} \right) w$$

It is worth highlighting that the labor market adjustment mechanism in an advanced capitalist economy with weak labor, which has been outlined in this subsection, is very different from the

[4] To keep the analysis simple, I do not distinguish between the total population and the labor force.

corresponding mechanism for an advanced capitalist economy with strong labor discussed in the previous subsection. In the latter, workers are powerful enough to attain their target wage share after technical change. This is made possible by collective action through labor unions and other such institutions and involves increases in the real wage rate to keep pace with labor productivity growth. The simple logic of demand and supply is supplanted in the labor market by collective action and wage bargaining. In the former, on the other hand, workers do not have the power to attain a target wage share. This is because the necessary supportive institutions are either missing or have been weakened by neoliberal assault. Instead, the real wage rate adjusts to clear the labor market, driven primarily by the logic of demand and supply.

To summarize, the classical-Marxian growth model for advanced capitalist economies with weak labor has five endogenous variables, w, c, g, r., and n, which are linked by five equations, (2), (3), (4), (10) and (12). To ensure meaningful equilibrium solutions of this model, I need the following restrictions.

Assumption 3. *The parameters* a_0, a_1, n_0, n_1 *are all strictly positive,* r_m *is weakly positive,* $0 < s_r < 1$*, and the reciprocal of capital productivity is bounded above*:

$$a_1 < a_1^{max} = \frac{s_r}{n_0 + s_r r_m} \tag{13}$$

This assumption is similar to assumption 2. It gives a lower bound for the capital productivity, $1/a_1$, that keeps the profit rate above the minimum threshold, r_m, so that we can rule out zero growth.

2.3. Meaningful equilibrium solutions

While describing various closures of the classical-Marxian model, I have imposed restrictions on parameters to ensure that meaningful solutions are guaranteed.[5] By meaningful equilibrium solutions, I mean the following: the equilibrium profit rate is positive and larger than the minimum required by capitalists to undertake positive investment:

$$r^* > r_m \geq 0 \tag{14}$$

and the equilibrium real wage rate is positive:

$$w^* > 0 \tag{15}$$

The former, in turn, ensures that the equilibrium growth rate is positive, $g^* > 0$, because:

$$g^* = s_r \left(r^* - r_m \right)$$

and the latter implies that consumption per worker is positive, $c^* > 0$, because, using (2) and (3), we see that:

$$c^* - w^* = \frac{a_1}{a_0} \left(r^* - g^* \right)$$

which, on using $g^* = s_r \left(r^* - r_m \right)$, becomes:

[5] Kaldor (1989) also imposes inequality restrictions to generate meaningful equilibrium solutions for some of his models.

$$c^* - w^* = \frac{a_1}{a_0}\left((1-s_r)r^* + s_r r_m\right) > 0$$

Proposition 1. *If assumption 1 holds, then the model of the labor surplus developing economy produces meaningful equilibrium solutions.*
Proposition 2. *If assumption 2 holds, then the model of the advanced capitalist economy with strong labor produces meaningful equilibrium solutions.*
Proposition 3. *If assumption 3 holds, then the model of the advanced capitalist economy with weak labor produces meaningful equilibrium solutions.*

Proofs of proposition 1, 2, and 3 are presented in the appendix.

3. Choice of technique, viability, and profitability

Suppose the economy starts in the long run equilibrium position with a technique of production given by (a_0, a_1). Now, suppose a new technique of production, given by (a_0', a_1'), becomes available; suppose further that this new technique of production satisfies assumption 1, 2, or 3, whichever is relevant.

3.1. The capitalist viability condition

Taking the existing equilibrium wage rate as given, we can use (2) to find the profit rate that a capitalist firm can *expect to* earn when the new technique of production is adopted as:

$$r^e = \frac{1 - a_0' w^*}{a_1'} \tag{16}$$

The capitalist firm, while evaluating the new technique of production, will compare the expected profit rate with the current equilibrium profit rate. It will adopt the new technique only if the expected profit rate exceeds the current equilibrium profit rate. Thus, the capitalist firm will adopt the new technique if $r^e > r^*$.

Let $\tilde{w}$ denote the *switch-point* real wage rate in the following sense: if $w^* > \tilde{w}$, then the new technique of production can be expected to increase the equilibrium profit rate. Since $r^e = \left(1 - a_0' w^*\right)/a_1'$ and $r^* = \left(1 - a_0 w^*\right)/a_1$, we see that:

$$\tilde{w} = \frac{a_1' - a_1}{a_0 a_1' - a_1 a_0'} \tag{17}$$

If $w^* > \tilde{w}$, then the capitalist firm will adopt the new technique of production and the economy will switch from an old to a new wage-profit frontier. Note that for a CU-LS technical change, we have $a_0 > a_0'$ and $a_1 < a_1'$; this ensures that $\tilde{w} > 0 : a_1' > a_1$ makes the numerator positive and $a_1/a_1' < 1 < a_0/a_0'$ makes the denominator positive.

3.2. Constant equilibrium profit rate

When technical change represented by the new technique of production, (a_0', a_1'), is adopted, let the equilibrium profit rate be denoted by r^{**}. If the LTFRP holds, that is, if the equilibrium profit rate falls after viable technical change, then we must have:

$$\frac{1 - a_0' w^{**}}{a_1'} \equiv r^{**} < r^* \equiv \frac{1 - a_0 w^*}{a_1}$$

which, using equality in the above relationship, gives us the constant equilibrium profit rate (CEPR) line:

$$w^{**} = \frac{a_1 - a_1'}{a_1 a_0'} + \left(\frac{a_0 a_1'}{a_0' a_1} \right) w^* \tag{18}$$

If the new equilibrium real wage rate is above the level given by (18), then the equilibrium profit rate falls after technical change.[6]

Figure 1 represents both the switch-point wage rate in (17) and the CEPR in (18) graphically. On the horizontal axis, we measure w^*, the existing equilibrium real wage rate, and on the vertical axis, we measure w^{**}, the equilibrium real wage rate after technical change. The (black) line OO' is the 45 -degree line through the origin. The (red) line AB represents the CEPR line.

Three things should be noted from figure 1. First, for a CU-LS technical change, we have $a_0 > a_0'$ and $a_1 < a_1'$; this implies that the intercept of the CEPR line, $(a_1 - a_1') / a_0' a_1$, is negative and it's slope, $(a_0 / a_0')(a_1' / a_1)$, is greater than 1. That is why the line AB starts below the origin and has a steeper slope than the 45 -degree line in the figure. Second, any point above (below) the line AB is associated with a fall (rise) in the equilibrium rate of profit with the new technique of production. Third, since our interest is only in viable technical change, we only need to consider initial equilibrium wage rates larger than $\tilde{w}$. Thus, we need to only consider the figure to the right of the dotted vertical line at $\tilde{w}$.[7]

4. Technical Change and Profitability

I am now ready to answer the main question of this article: when will the equilibrium rate of profit fall after viable, CU-LS technical change?

4.1. Labor surplus economy

The closure for the labor surplus economy is given by $w = \bar{w}$, that is, the real wage rate remains constant. This implies that the equilibrium real wage rate remains unchanged before and after viable technical change, that is, $w^{**} = w^*$. Thus, for any $w^* > \tilde{w}$, the economy will lie on the segment HO' of the 45-degree line after viable technical change. Since HO' lies entirely below HB, the equilibrium profit rate will unambiguously rise after viable technical change.

4.2. Advanced capitalist economy with strong labor

The closure for the advanced capitalist economy with strong labor is given by $\pi = \bar{\pi}$; that is, the profit share remains constant. Hence, $1 - a_0 w^*$, which is the profit share before technical change, is equal to $1 - a_0' w^{**}$, which is the profit share after technical change. This implies that:

$$w^{**} = \left(\frac{a_0}{a_0'} \right) w^* \tag{19}$$

In figure 1, (19) is depicted by the (blue) line OC through the origin. The line OC represents the equilibrium real wage rate after technical change as a function of the equilibrium real wage rate before technical change. Hence, we can call it the *equilibrium wage-wage curve* (EWWC).

[6] The ratio w^{**} / w^* generalizes what Basu (2021: 286) has called the Marx-Okishio threshold.

[7] The fact that AB intersects that 45-degree line at $w^* = \tilde{w}$ can be verified by setting $w^* = w^{**} = \tilde{w}$ in (18), solving for $\tilde{w}$, and then comparing with (17).

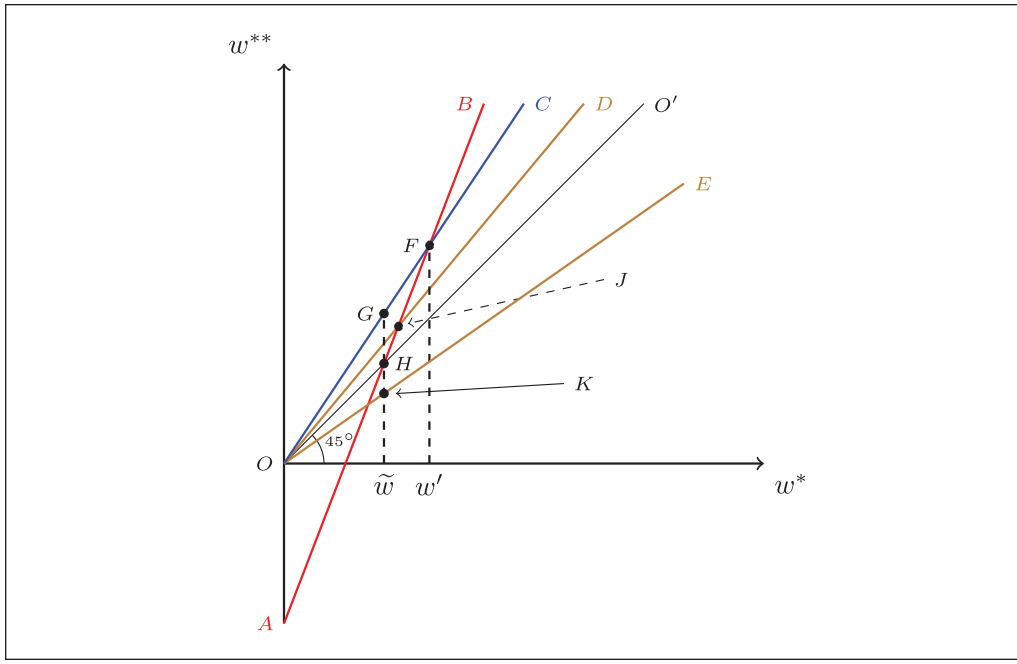


Figure 1. Viable, CU-LS technical change, and profitability: w^* and w^{**} denote the equilibrium real wage rates before and after technical change, respectively. The (black) line OO' is the 45-degree line through the origin; it represents a labor surplus economy. The (blue) line OC through the origin represents an advanced capitalist economy with strong labor. The (brown) lines, OD and OE, through the origin represent an advanced capitalist economy with weak labor. The (red) line AB represents the CEPR curve. Points above (below) AB correspond to a fall (rise) in the equilibrium profit rate after technical change.

Note that the slope of OC, a_0 / a_0', is lower than the slope for AB, $\left(a_0 / a_0'\right)\left(a_1' / a_1\right)$, because $a_1' > a_1$ (CU-LS technical change). We also see from the figure that the CEPR line, AB, and the EWWC for the advanced capitalist economy with strong labor, OC, intersect at F. Let us denote the corresponding level of the real wage rate by w'.

We can now see what happens to the equilibrium profit rate after viable, CU-LS technical change. If the initial (equilibrium) real wage rate lies in the open interval $\left(\tilde{w}, w'\right)$, then the equilibrium profit rate falls after viable, CU-LS technical change. This is because the segment GF of OC lies completely above the corresponding segment HF of AB. But if the initial (equilibrium) real wage rate is larger than w', then the equilibrium profit rate rises after viable, CU-LS technical change. This is because the segment FC of OC lies completely below the corresponding segment FB of AB. But can the real wage rate ever be larger than w'?

Using the equations for AB and OC, we can calculate that $w' = 1 / a_0$. This is not feasible because at this real wage rate, the corresponding profit rate will be 0. But recall that the profit rate is bounded below by r_m, that is, $r \geq r_m > 0$. Hence, $w \leq \left(1 / a_0\right) - \left(a_1 / a_0\right) r_m < \left(1 / a_0\right) = w'$. Thus, the initial equilibrium real wage rate for the economy always resides in the open interval $\left(\tilde{w}, w'\right)$. Hence, now we can state that for the model of the advanced capitalist economy with strong labor, any viable CU-LS technical change will unambiguously lead to a fall in the equilibrium rate of profit. Marx's LTFRP reigns supreme in an advanced capitalist economy with strong labor.

4.3. Advanced capitalist economy with weak labor

For the advanced capitalist economy with weak labor, the analysis is more involved. From the model equations in section 2.2.3, we can solve for the equilibrium real wage rate before technical change as:

$$w^{*} = \frac{s_r - a_1 n_0 - a_1 s_r r_m}{a_0 s_r + a_1 n_1} \tag{20}$$

and the equilibrium real wage rate after viable technical change as:

$$w^{**} = \frac{s_r - a_1' n_0 - a_1' s_r r_m}{a_0' s_r + a_1' n_1} \tag{21}$$

To address the question about the possibility of the LTFRP in this economy, we need the following result.

Proposition 4. *For the advanced capitalist economy with weak labor, we have*:

$$\frac{w^{**}}{w^{*}} < \frac{a_0}{a_0'} \tag{22}$$

The details of the proof are presented in the appendix. Here, let us think about its implication and intuition. With reference to figure 1, the EWWC for an advanced capitalist economy with weak labor will pass through the origin and will have a (positive) slope that is strictly less than the slope of the CEPR, AB. Of course, the implication of viable, CU-LS technical change on the equilibrium rate of profit depends on whether the slope of the EWWC is larger or smaller than unity. To discuss the details of these two subcases involving the slope of the EWWC for an advanced capitalist economy with weak labor, I need the following inequality that was demonstrated to be true in the proof of proposition 4 (in the appendix):

$$\frac{\dfrac{s_r}{a_1'} - \left(n_0 + s_r r_m\right)}{\dfrac{s_r}{a_1} - \left(n_0 + s_r r_m\right)} < \frac{\dfrac{s_r}{a_1'} + \dfrac{n_1}{a_0'}}{\dfrac{s_r}{a_1} + \dfrac{n_1}{a_0}} \tag{23}$$

Now consider two subcases.

Case 1. Suppose the labor-saving aspect of the new CU-LS technique of production is relatively weak; that is, a_0' / a_0 is not too small because a_0' differs from a_0 by a small amount. To be more precise, if multiplying the right-hand side of (23) by a_0' / a_0 does *not* disturb the inequality, we have:

$$\frac{\dfrac{s_r}{a_1'} - \left(n_0 + s_r r_m\right)}{\dfrac{s_r}{a_1} - \left(n_0 + s_r r_m\right)} < \left(\frac{a_0'}{a_0}\right)\left(\frac{\dfrac{s_r}{a_1'} + \dfrac{n_1}{a_0'}}{\dfrac{s_r}{a_1} + \dfrac{n_1}{a_0}}\right) \tag{24}$$

then the equilibrium rate of profit *will* rise after viable, CU-LS technical change.

Case 2. Suppose the labor-saving aspect of the new CU-LS technique of production is relatively strong; that is, a_0'/a_0 is quite small (because a_0' differs from a_0 by a large amount) such that multiplying the right-hand side of (23) by $0 < a_0'/a_0 < 1$ reverses the inequality. Thus, we have:

$$\left(\frac{a_0'}{a_0}\right)\left(\frac{\frac{s_r}{a_1'}+\frac{n_1}{a_0'}}{\frac{s_r}{a_1}+\frac{n_1}{a_0}}\right) < \frac{\frac{s_r}{a_1'}-\left(n_0+s_r r_m\right)}{\frac{s_r}{a_1}-\left(n_0+s_r r_m\right)} \tag{25}$$

in such a situation, the equilibrium rate of profit *can* fall after viable, CU-LS technical change, if the initial equilibrium real wage rate was not too high to begin with.

The key to understanding the above two subcases is to work with the condition in (24). If this condition is satisfied and we are in case 1, then $w^{**}/w^* < 1 < a_0/a_0'$. To see this, simplify the right-hand side of (24) to get:

$$\frac{\frac{s_r}{a_1'}-\left(n_0+s_r r_m\right)}{\frac{s_r}{a_1}-\left(n_0+s_r r_m\right)} < \frac{\frac{a_0' s_r}{a_1'}+n_1}{\frac{a_0 s_r}{a_1}+n_1}$$

now multiply both sides by a_1'/a_1 and rearrange, and using (20) and (21) get the result that $w^{**}/w^* < 1 < a_0/a_0'$.

With this result, we can now refer back to figure 1 and see that, under case 1 (where labor productivity growth is small), any EWWC for the advanced capitalist economy with weak labor, OE, lies completely below OO'. Hence, for any initial equilibrium real wage rate higher than the switch-point wage rate, $\tilde{w}$, the segment of the EWWC, KE, lies completely below the segment EB of the CEPR. Hence, the equilibrium profit rate will rise after technical change.

On the other hand, if the condition in (24) is violated and we are under case 2 (where labor productivity growth is strong), then we have $1 < w^{**}/w^* < a_0/a_0'$. In this case, as we can see from figure 1, any EWCC for the advanced capitalist economy with weak labor, OD, lies between OC and OO'. It intersects the CEPR line at J. Let us denote the corresponding level of w^* (equilibrium real wage rate before technical change) as w'' (not shown in the figure to prevent clutter). Now, using figure 1, we see the following: if w^* lies in the interval $\left(\tilde{w}, w''\right)$, then the equilibrium profit rate will fall after viable, CU-LS technical change; if $w'' < w^* < w'$, then the equilibrium profit rate will rise after viable, CU-LS technical change.

We saw above that within the class of viable, CU-LS technical change, a relatively weak labor saving component (case 1) leads to a rise and a relatively strong labor saving component (case 2) can lead to a fall, in the equilibrium rate of profit in an advanced capitalist economy with weak labor. What is the *intuition* for these results?

We can use figure 2, which depicts the determination of the equilibrium real wage rate in the model of the advanced capitalist economy with weak labor, to build some intuition. In figure 2, the downward sloping line, AA', captures the accumulation function in (4) written in terms of the real wage rate (where we substitute out the profit rate using (2)):

$$g = \left(\frac{s_r}{a_1} - s_r r_m\right) - \left(\frac{s_r a_0}{a_1}\right) w \tag{26}$$

Written this way, the accumulation function can be seen as a labor demand function. The vertical and horizontal intercepts of this labor demand function are given by $\left(s_r/a_1\right) - s_r r_m$ and $\left(1 - a_1 r_m\right)/a_0$, respectively.

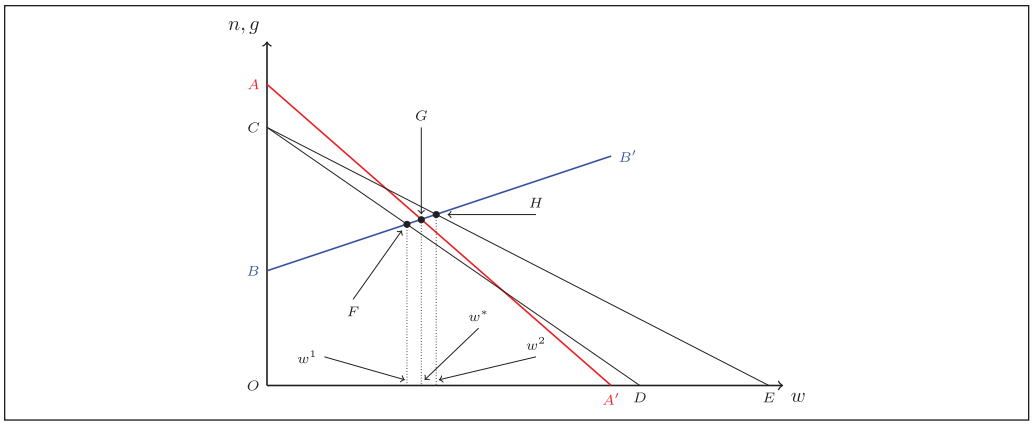


Figure 2. Viable, CU-LS technical change and profitability in an advanced capitalist economy with weak labor: *w*, *n* and *g* denote, respectively, the real wage rate, the labor supply growth rate and the accumulation rate. BB' is the labor supply function; AA' is the accumulation function before technical change. *CD* and *CE* represent the accumulation function after technical change for two different scenarios.

In figure 2, the upward sloping line, BB', depicts the labor supply function in (10):

$$n = n_0 + n_1 w$$

The point of intersection of the labor supply and labor demand functions *G* at depicts the equilibrium:

$$g = n$$

with w^* the equilibrium real wage rate.

Now consider two different cases of CU-LS technical change keeping the change in a_1 the *same in both cases*. In the first case, a_0 falls by a small amount (this corresponds to case 1 above); in the second case, a_0 falls by a large amount (this corresponds to case 2 above). The labor demand functions in the first and second cases are depicted by *CD* and *CE*, respectively. They share the same vertical intercept because they have the same a_1' (after technical change), but the size of the horizontal intercept, $\left(1-a_1 r_m\right)/a_0'$, is larger in the second case because a_0 falls by a larger amount in case 2. The equilibrium real wage rates after technical change in case 1 and case 2, respectively, are w^1 and w^2. From figure 2, we see that $w^1 < w^* < w^2$. What is driving this result?

The labor supply function does not change with technical change. This is because the parameters of the labor supply function, n_0 and n_1, are not impacted by the choice of technique. Hence, in figure 2, BB' remains unchanged after technical change. On the other hand, the labor demand function in (26) does change with technical change. In fact, with CU-LS technical change, two things about the labor demand function change: the slope (capturing the responsiveness of labor demand to changes in the real wage rate) declines in magnitude; that is, AA' becomes flatter and the vertical intercept (capturing the maximum possible accumulation rate) declines. Two possible configurations of such changes are depicted in figure 2.

In the first case (case 1), the labor demand curve shifts to *CD* after technical change. At the original equilibrium real wage rate, w^*, there is an excess supply of labor. Hence, the real wage

rate falls to equate demand and supply. As a result, the new equilibrium real wage rate is lower and the new equilibrium profit rate is higher after viable technical change. In the second case (case 2), the labor demand curve shifts to *CE* after technical change. At the original equilibrium real wage rate, w^*, there is an excess demand for labor. Hence, the real wage rate rises to equate demand and supply. As a result, the new equilibrium real wage rate is higher and the profit rate *can* be lower after viable technical change.

5. Conclusion

Technical change in capitalist economies consist of at least two stages. In the first stage, new techniques of production are generated by R&D efforts of public and private institutions. In the second stage, capitalist firms evaluate whether a new technique will increase their rate of profit at the existing real wage rate. They adopt the new technique only if it is expected to increase their rate of profit. Such techniques of production can be called capitalistically viable.

Neither are all capitalistically viable techniques socially beneficial, nor are all socially desirable techniques capitalistically viable (Foley 1986; Basu 2021). But leaving this issue aside, we can ask whether viable technical change might ever lead to a fall in the equilibrium rate of profit. If this were to happen then it would paradoxically undermine capitalists: even when they only adopt techniques of production that are expected to raise the rate of profit, they end up in the new long-run equilibrium with a lower profit rate!

In this article, I have investigated this question in classical-Marxian models of economic growth with alternative labor market closures. These closures map onto three distinct structural and institutional settings of capitalist economies: a labor surplus developing economy (where the real wage rate remains fixed), an advanced capitalist economy where labor has relatively strong bargaining power vis-à-vis capital (where the wage share remains fixed), and finally, an advanced capitalist economy with weak labor (where the unemployment remains fixed).

Whether the equilibrium profit rate falls after viable technical change depends crucially on the labor market closure, a point that was highlighted almost a century ago by Dobb (1945). In particular, I show that the movement of the equilibrium profit rate after viable, CU-LS technical change is diametrically opposite in labor surplus economies versus in advanced capitalist economies with strong labor. In the former, the equilibrium profit rate will rise, and in the latter, it will fall, after viable, CU-LS technical change. On the other hand, both possibilities are open in an advanced capitalist economy with weak labor. If the labor-saving aspect of viable, CU-LS technical change is weak, that is, labor productivity growth is small, then the equilibrium profit rate will rise; but if the labor-saving aspect of viable, CU-LS technical change is relatively strong, that is, labor productivity growth is high, then the equilibrium profit rate can fall.

Through the analysis of this article we have seen that the only context where Marx's LTFRP gains traction is in an advanced capitalist economy with strong labor. Moreover, the specific way in which I have captured strong labor in an advanced capitalist economy is by keeping the wage share of national income fixed—this is the classical conventional wage share model of Foley et al. (2019). Thus, labor is strong in the specific sense that the real wage rate grows in tandem with labor productivity, and this is the precise context in which the LTFRP becomes possible for viable, CU-LS technical change.

I would like to conclude with two comments. First, while the reasons for the emergence and recurrence of MBTC have been studied within the induced innovation literature (Foley et al. 2019: chapter 7) and within game-theoretic modeling frameworks (Baldani and Michl 2000; Kang and Rieu 2009), the question about the constancy of the wage share has attracted much less attention. What plausible political and economic mechanisms might ensure that the wage share remains stable over time? How best to model such processes? In this article, I have offered a

simple model of working-class collective action that results in a constant wage share in the long run. More work is needed to deepen this and other such models as we pursue what Tom Michl has aptly called the "falling rate of profit research program" (Michl 2023).

Second, some scholars and activists have sought to portray Marx's theory of capitalist crisis as deriving *solely* from the LTFRP presented in volume 3 of *Capital*. The theoretical analysis in this article suggests that tethering a theory of capitalist crisis solely to the LTFRP might be an unduly narrow way to interpret Marx's ideas. A careful reading of Marx's dispersed writings on crisis tendencies in capitalism suggests that he might have had other mechanisms in mind too (Basu 2019). Later scholarship does not suggest that the LTFRP is not important. It certainly is an important mechanism of crisis and it remains so. But it is not the only one. In different labor market setups, it is possible for the LTFRP to not operate, as this article seeks to demonstrate, and yet there are certainly other mechanisms that can give rise to crisis tendencies. Marxist scholarship will benefit from not limiting the ambit of theorizing about crisis tendencies in capitalism to a "one size fits all" LTFRP mechanism.

Appendices

A. Appendix: Proofs

A.1. Equilibrium profit rate. For each of the models described above, we can solve for the equilibrium profit rate, r^*. For the labor surplus developing economy, we have:

$$r^* = \frac{1 - a_0 \bar{w}}{a_1} \tag{27}$$

Thus, the equilibrium profit rate is impacted by the technological parameters, a_0 and a_1, and the exogenous real wage rate, $\bar{w}$.

For the advanced capitalist country with strong labor, we have:

$$r^* = \frac{\bar{\pi}}{a_1} \tag{28}$$

so that the equilibrium profit rate is impacted by the single technological parameter a_1 and the exogenous profit share, $\bar{\pi}$.

For the advanced capitalist economy with weak labor, we have:

$$r^* = \frac{a_0 \left(n_0 + s_r r_m \right) + n_1}{a_0 s_r + a_1 n_1} \tag{29}$$

so that the equilibrium profit rate is impacted by the accumulation propensity, s_r, the minimum threshold profit rate for positive investment, r_m, the parameters of the labor supply growth function, n_0 and n_1, and most crucially for the analysis of this article, on the technology parameters, a_0 (reciprocal of labor productivity) and a_1 (reciprocal of capital productivity).

A.2. Equilibrium values of other endogenous variables. The equilibrium values of the other four endogenous variables are not directly relevant for the analysis presented in this article. But they can be easily calculated using the equations of the relevant model and the expression for the relevant equilibrium profit rate:

$$g^* = s_r \left(r^* - r_m \right) \tag{30}$$

$$w^* = \frac{1}{a_0} - \frac{a_1}{a_0} r^* \tag{31}$$

$$c^* = \frac{1}{a_0} - \frac{a_1}{a_0} g^* \tag{32}$$

A.3. Proof of proposition 1. Since assumption 1 holds, we have:

$$\frac{1 - a_0 \overline{w}}{r_m} > a_1$$

which is equivalent to:

$$\frac{1 - a_0 \overline{w}}{a_1} > r_m$$

because $a_1 > 0$ and $r_m > 0$. Using the definition of the equilibrium profit rate, r^* in (27), this is equivalent to:

$$r^* > r_m \geq 0$$

Moreover, we know that $w^* = \overline{w} > 0$ by assumption.

A.4. Proof of proposition 2. Using assumption 2 and the expression for the equilibrium profit rate in (28), we have:

$$r^* = \frac{\overline{\pi}}{a_1} > r_m$$

because $a_1 > 0$ and $r_m > 0$. Moreover, we have:

$$w^* = \frac{1 - \overline{\pi}}{a_0} > 0$$

because $0 < \overline{\pi} < 1$ and $a_0 > 0$.

A.5. Proof of proposition 3

A.5.1 Two lemmas.

Lemma 1. $r^* > r_m$ is equivalent to $a_1 n_1 r_m < n_1 + a_0 n_0$.

Proof. Using the expression for the equilibrium rate of profit in (29), $r^* > r_m$ is equivalent to:

$$\frac{a_0 \left(n_0 + s_r r_m \right) + n_1}{a_0 s_r + a_1 n_1} > r_m$$

which is equivalent to:

$$a_0 n_0 + a_0 s_r r_m + n_1 > a_0 s_r r_m + a_1 n_1 r_m$$

because $a_0 s_r + a_1 n_1 > 0$. Canceling terms on both sides, the above is equivalent to:

$$a_1 n_1 r_m < n_1 + a_0 n_0$$

which establishes the result.

Lemma 2. $w^* > 0$ is equivalent to $r_m < (1/a_1) - (n_0 / s_r)$.

Proof. Since $w^* = (1 - a_1 r^*)/a_0$:

$$w^* > 0 \Leftrightarrow r^* < 1/a_1$$

Using the expression for the equilibrium rate of profit in (29), this is equivalent to:

$$\frac{a_0 (n_0 + s_r r_m) + n_1}{a_0 s_r + a_1 n_1} < \frac{1}{a_1}$$

which is equivalent to:

$$a_1 a_0 n_0 + a_1 a_0 s_r r_m + a_1 n_1 < a_0 s_r + a_1 n_1$$

because $a_0 s_r + a_1 n_1 > 0$. Canceling terms on both sides, dividing through by $a_0 a_1 s_r$ and rearranging, the above is equivalent to:

$$r_m < \frac{1}{a_1} - \frac{n_0}{s_r}$$

because $a_0 > 0, a_1 > 0, s_r > 0$.

A.5.2 Proof of proposition 3. Since assumption 3 holds, we have:

$$a_1 < \frac{s_r}{n_0 + s_r r_m}$$

Since $n_0 + s_r r_m > 0, s_r > 0, a_1 > 0$, this is equivalent to:

$$\frac{n_0 + s_r r_m}{s_r} < \frac{1}{a_1}$$

which, in turn, is equivalent to:

$$r_m < \frac{1}{a_1} - \frac{n_0}{s_r}$$

Now using lemma 2, we get $w^* > 0$.

Since:

$$r_m < \frac{1}{a_1} - \frac{n_0}{s_r}$$

multiplying through by $a_1 n_1 \geq 0$, gives:

$$a_1 n_1 r_m \leq n_1 - \frac{a_1 n_1 n_0}{s_r}$$

which implies that:

$$a_1 n_1 r_m \leq n_1 - \frac{a_1 n_1 n_0}{s_r} \leq n_1 < n_1 + a_0 n_0$$

because $\left(a_0 a_1 n_1\right)/s_r \geq 0$ and $a_0 n_0 > 0$. Now using lemma 1, we get $r^* > r_m$.

A.6. Proof of proposition 4. I use the following result for the proof: if $0 < x < a < b$, $y > 0$ and $z > 1$, then:

$$\frac{a-x}{b-x} < \frac{a}{b} < \frac{a+y}{b+y} < \frac{a+yz}{b+y} \tag{33}$$

The rightmost inequality in (33) is obvious because $yz > y$. To see the next inequality from the right, start with $by > ay$ and add ab to both sides to get $b\left(a+y\right) > a\left(b+y\right)$. Now dividing both sides by $\left(b+y\right)b > 0$ gives the result. To see the first equality, start with $-bx < -ax$ and add ab to both sides to get $b\left(a-x\right) < a\left(b-x\right)$. Now dividing both sides by $b\left(b-x\right) > 0$ gives the result.

Now, let:

$$a = \frac{s_r}{a_1'}, b = \frac{s_r}{a_1}, x = n_0 + s_r r_m, y = \frac{n_1}{a_0}, z = \frac{a_0}{a_0'}$$

and with these values, note that $0 < a < b$ because $a_1' > a_1$ (CU-LS technical change), $0 < x < a$ because $w^{**} > 0$ in (21), $y > 0$ because $n_1, a_0 > 0$ and $z > 1$ because $0 < a_0' < a_0$ (CU-LS technical change). Thus, we can use (33) to get:

$$\frac{\frac{s_r}{a_1'} - \left(n_0 + s_r r_m\right)}{\frac{s_r}{a_1} - \left(n_0 + s_r r_m\right)} < \frac{\frac{s_r}{a_1'}}{\frac{s_r}{a_1}} < \frac{\frac{s_r}{a_1'} + \frac{n_1}{a_0}}{\frac{s_r}{a_1} + \frac{n_1}{a_0}} < \frac{\frac{s_r}{a_1'} + \frac{n_1}{a_0}\frac{a_0}{a_0'}}{\frac{s_r}{a_1} + \frac{n_1}{a_0}}$$

For us, the two ends of the above chain of inequalities is most relevant:

$$\frac{\frac{s_r}{a_1'} - \left(n_0 + s_r r_m\right)}{\frac{s_r}{a_1} - \left(n_0 + s_r r_m\right)} < \frac{\frac{s_r}{a_1'} + \frac{n_1}{a_0}\frac{a_0}{a_0'}}{\frac{s_r}{a_1} + \frac{n_1}{a_0}} = \frac{\frac{s_r}{a_1'} + \frac{n_1}{a_0'}}{\frac{s_r}{a_1} + \frac{n_1}{a_0}} \tag{34}$$

where the right-hand side can be simplified to:

$$\frac{\frac{s_r}{a_1'} + \frac{n_1}{a_0'}}{\frac{s_r}{a_1} + \frac{n_1}{a_0}} = \frac{a_1}{a_1'}\left(\frac{a_0 a_0' s_r + a_0 a_1' n_1}{a_0 a_0' s_r + a_0' a_1 n_1}\right)$$

so that (34) becomes:

$$\frac{\frac{s_r}{a_1'}-\left(n_0+s_r r_m\right)}{\frac{s_r}{a_1}-\left(n_0+s_r r_m\right)}<\frac{a_1}{a_1'}\left(\frac{a_0 a_0' s_r+a_0 a_1' n_1}{a_0 a_0' s_r+a_0' a_1 n_1}\right)$$

Now multiplying both sides by $a_1'/a_1>0$ and simplifying, we get:

$$\frac{s_r-a_1'\left(n_0+s_r r_m\right)}{s_r-a_1\left(n_0+s_r r_m\right)}<\left(\frac{a_0}{a_0'}\right)\left(\frac{a_0' s_r+a_1' n_1}{a_0 s_r+a_1 n_1}\right)$$

which can be rearranged to give:

$$\frac{\left[s_r-a_1'\left(n_0+s_r r_m\right)\right]\left[a_0 s_r+a_1 n_1\right]}{\left[s_r-a_1\left(n_0+s_r r_m\right)\right]\left[a_0' s_r+a_1' n_1\right]}<\frac{a_0}{a_0'}$$

Using (20) and (21), the left-hand side is seen to be w^{**}/w^{*}, so that the proof is complete.

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